

Auditable Flourishing v6.2 Measurement and Construct Handbook

Status and role

This handbook is the binding implementation companion for measurement and construct claims made under Auditable Flourishing v6.2. Use of the F2PHS/F2PHS+ vocabulary is optional; satisfaction of the applicable measurement-integrity functions is not. The handbook does not define a universal theory of flourishing. It provides a default vocabulary, construct documentation rules, measurement-card fields, distribution and invariance safeguards, and anti-Goodhart controls for candidates that use indicators or structured evidence.

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1. Functional coverage and ontology pluralism

Candidates must address protected agency; fairness and non-domination; material and capability security; peaceful pluralism and social integrity; and ecological and intergenerational continuity, or provide a principled explanation of non-equivalence. F2PHS is the default public vocabulary, not a mandatory ontology.

A two-way crosswalk is required:

1. candidate construct -> protocol function; and
2. protocol function -> candidate construct, integrated location, or justified omission.

Reviewers must permit principled non-equivalence. A candidate may explain that a function is relational, embedded, or not separable in its ontology. The question is whether protection and evidence are materially present, not whether labels match.

1.1 Operational anchor discipline

Every construct or Stage B dimension must declare: object and unit of analysis; protected-floor relation; applicable and non-applicable domains; observable evidence; category anchors; counterexamples; uncertainty; missingness; subgroup and distribution implications; evidence maturity; reviewer confidence; gaming pathways; update trigger; retirement condition; and authority boundary. An ordinal label is not a unit. Categories may be ordered only through defended descriptors and may not be summed, averaged, differenced, or assigned decimal weights without a validated measurement model.

A positive category requires evidence that distinguishes the category from adjacent alternatives. Unknown, not applicable, and not reviewed remain distinct. Applicability cannot be chosen strategically to improve a profile. A category that masks a material C0-C5 defect triggers Stage A re-entry rather than remaining a weak compensable score.

2. Functional architecture

2.1 Harmony is not conformity

Harmony is the most culturally vulnerable term in F2PHS, so it requires a strict definition. In this protocol, harmony means low violence, protected dissent, constructive pluralism, conflict-resolution capacity, social trust, meaningful relationships, and institutional legitimacy. It does not mean unanimity, obedience, suppression of contestation, quietism, or absence of disagreement. A society can be harmonious only if dissent remains protected, grievances can be voiced, and conflict can be processed without coercion, dehumanization, or organized violence. This anti-conformity rule is part of the definition rather than a later caveat: if dissent, complaint, minority expression, or lawful opposition is suppressed, the Harmony dimension fails its integrity condition even if reported trust or visible order appears high.

Operationally, protected dissent and constructive pluralism should be measured through portfolios rather than a single trust metric. Candidate indicators may include independent measures of civil liberties, judicial independence, protest-rights protection, journalist or opposition detention, protest-policing harm, complaint-channel accessibility, deliberative-democracy indicators, mediation access, and polarization or dehumanization measures. These indicators should be interpreted together so that high reported trust or low visible conflict cannot mask coercion, censorship, or fear-induced silence.

2.2 Indicator taxonomy

Every F2PHS property must be represented by an indicator portfolio. A portfolio must include outcome indicators, process indicators, integrity indicators, and early-warning indicators. This distinction is essential because a society can improve an outcome measure while degrading process fairness, manipulate a reporting system while appearing compliant, or fail to detect early-warning signs before slower outcome indicators shift.

Indicator taxonomy matrix: one portfolio per F2PHS property

Property	Outcome	Process	Integrity	Early warning
Free	civil liberties	redress time	complaint suppression	emergency decrees
Fair	appeal parity	court access	data gaps	subgroup divergence
Prosperous	poverty gap	service coverage	survey capture	health-cost spike
Harmonious	violence rate	mediation access	bot manipulation	polarization
Sustainable	emissions	EIA compliance	greenwashing	water stress

Figure 1. Indicator taxonomy matrix. Each property requires outcome, process, integrity, and early-warning indicators.

Table 1. Indicator taxonomy.

Indicator type	Question answered	Example	Why it matters
Outcome	What is the actual condition?	Poverty headcount, avoidable mortality, homicide, emissions trajectory, life satisfaction	Shows whether the target state is improving or declining
Process	Is governance operating correctly?	Appeal timeliness, procurement review, procedural safeguard compliance, participatory quality	Prevents outcome-only shortcuts and compliance theater
Integrity	Is the measurement system being gamed?	Missingness spikes, complaint suppression, sudden definition changes, greenwashing audits	Detects manipulation, proxy drift, and reporting capture
Early warning	What signals emerging failure before collapse?	Food-price spikes, appeal backlogs, groundwater depletion, trust collapse, heat mortality trends	Supports prevention rather than post hoc repair

2.3 Suggested core factor map

Table 2. Suggested core factor map.

Factor	Why it belongs	Example indicators	Evidence strength	Main measurement risk
Freedom and agency	Central to rights and capabilities	Civil liberties, privacy, association, voice, appeal access	High	Rights-on-paper versus lived reality
Fairness and justice	Required for legitimacy and non-domination	Due-process access, appeal outcomes, discrimination gaps, corruption control	High	Selective enforcement and underreporting
Health	Universal flourishing component	Life expectancy, HALE, avoidable mortality, mental distress, catastrophic health spending	Very high	Lagging data and mental health comparability
Material security	Basic condition for stable life	Poverty rate and gap, food insecurity, housing insecurity, debt stress	Very high	Hidden deprivation and informal economies
Education and knowledge	Capability expansion and resilience	Literacy, numeracy, completion, learning outcomes, digital access	High	Schooling substituted for learning
Safety and peace	Violence collapses flourishing	Homicide, victimization, conflict incidence, displacement	Very high	Authoritarian order masking repression
Social connection and trust	Supports cooperation and wellbeing	Loneliness, social support, interpersonal trust, institutional trust	Moderate-high	Survey manipulation and cultural variance
Institutional legitimacy	Enables compliance without coercion	Rule of law, procedural fairness, trust in courts, state capacity	Moderate-high	Halo effects and politicization
Ecological integrity	Life-support condition	GHG trajectory, PM2.5, water stress, land use, biodiversity proxies	Very high	Offset theater and boundary mismatch
Intergenerational resilience	Flourishing must persist	Adaptation capacity, critical infrastructure resilience, disaster preparedness, tail-risk exposure	Moderate-high	Modeled uncertainty and long time horizons
Meaning and subjective wellbeing	Captures lived quality not visible in material statistics	Life satisfaction, purpose, affect balance, character, close relationships	Moderate-high	Adaptation, response bias, suppression of dissent

Subjective wellbeing and meaning are core field dimensions because lived experience matters and cannot be inferred from material statistics alone. However, subjective wellbeing and meaning cannot override rights floors, ecological boundaries, protected dissent, or independently evidenced harms. They operate as part of the

flourishing field and as validation signals, not as permissions to ignore coercion, ecological damage, or subgroup injury.

2.4 Ecological ceilings and planetary boundaries

The sustainability horizon should not be operationalized only through emissions or generic environmental indicators. A credible protocol must engage the planetary-boundaries literature. The 2025 Planetary Health Check reports that seven of nine boundaries have been breached and that all breached boundaries are under increasing pressure (Sakschewski et al., 2025). Auditable Flourishing therefore treats planetary boundaries as a canonical ecological reference set for any framework claiming civilizational scope. Candidate frameworks may use alternative ecological models, but they must justify why those models are at least as protective, measurable, and updateable.

Local ecological conditions that directly bear on life, health, subsistence, cultural continuity, or protected community survival may instantiate floor-level protections. Planetary-boundary conditions and long-horizon resilience constraints belong to the horizon layer. This clarifies why ecological integrity can appear both as a floor where immediate rights or subsistence are implicated and as a horizon where long-term biospheric stability is at stake.

This horizon framing is also grounded in the social-foundation and ecological-ceiling tradition and the planetary-boundaries literature (Raworth, 2017; Rockström et al., 2009; Steffen et al., 2015; Richardson et al., 2023), while the protocol keeps those sources subordinate to the open-core, rights, audit, and update requirements defined here.

Table 3. Planetary boundary reference families for the sustainability horizon.

Boundary family	Protocol use	Measurement caution
Climate change	Hard horizon constraint; track emissions, temperature anomalies, and carbon-budget alignment	Avoid offset theater and delayed-action accounting
Biosphere integrity	Species, habitat, and ecosystem function indicators	Do not let aggregate land protection hide local ecosystem collapse
Land-system change	Land conversion, deforestation, and degradation	Distinguish legal land-use change from ecological function
Freshwater change	Surface and groundwater stress, flow alteration	Monitor local depletion and transboundary effects
Biogeochemical flows	Nitrogen and phosphorus loading	Avoid reporting only national totals when watershed harm is local
Ocean acidification	Ocean chemistry and marine ecosystem risk	Track lagged effects and marine food-system exposure
Novel entities	Persistent pollutants, plastics, synthetic chemicals, AI/tech-adjacent material streams where relevant	Measurement gaps are high; uncertainty should trigger precaution
Stratospheric ozone	Monitor recovery and policy compliance	Do not treat recovery as evidence that all boundaries are safe

Boundary family	Protocol use	Measurement caution
Atmospheric aerosol loading	Regional air pollution and radiative forcing	Strong regional inequality and health burden effects

2.5 F2PHS+ dictionary status, evidence hierarchy, safe-and-just boundaries, and construct-validity guardrails

F2PHS is the top-level communication layer. F2PHS+ is the expanded subdimensional reference dictionary; together with measurement cards, rights/horizon/integrity constraints, and the challenge protocol constitute the operational method. Reviewers should therefore reject submissions that merely rename existing indicators into the five F2PHS labels without showing construct links, evidence quality, subgroup behavior, boundary relevance, and update triggers.

For ecological and intergenerational claims, the protocol distinguishes global horizon indicators from local floor harms. Planetary-boundary and safe-and-just Earth-system-boundary work can guide horizon assessment (Rockström et al., 2023), but a local watershed, air-quality, land, food, or subsistence harm may also trigger a floor-level rights analysis. Boundary metrics are uncertain and sometimes politically contested; civilizational-scope claims should therefore test robustness across plausible ecological models when feasible.

Table 4. Evidence hierarchy distinguishes legal authority, peer-reviewed constructs, official statistics, candidate artifacts, and synthetic evidence.

Evidence tier	Allowed use	Reviewer caution
Primary law / binding authority	Defines legal duties, affected-party rights, authorization limits, and mandatory impact-assessment obligations.	Do not treat protocol admissibility as legal permission.
Peer-reviewed construct or method literature	Supports construct validity, causal reasoning, reliability, measurement invariance, or ecological boundary logic.	Prefer for load-bearing scientific claims.
Official institutional statistics or reports	Supports current contextual facts and indicator selection.	Verify edition/date; do not make reports carry foundational theory.
Candidate-provided artifacts	Supports Stage A/Stage B evaluation only when independently replayable or challengeable.	Disclosure alone is not evidence of real-world validity.
Synthetic data or illustrative scenarios	Supports L1–L2 replay and reviewer training.	Cannot support deployment or superiority claims.

Version 6.2 treats F2PHS+ as a provisional canonical dictionary for challenge rounds, not as a final theory of human or civilizational value. Its function is reviewer discipline: each submitted framework must show which floor, field, horizon, and integrity constructs it uses, which indicators serve those constructs, what those indicators miss, and what would count as construct drift or proxy failure.

Minimum dictionary rule. Each F2PHS+ property must disclose six elements: protected floor, constructive field, long-horizon risk, integrity safeguard, prohibited misuse, and example measurement-card families. A candidate framework may use different labels, but it must map its terms to these elements or explain why an alternative dictionary is at least as protective, inspectable, and updateable.

Construct-validity guardrail. Indicator selection should be treated as a validity argument rather than a dashboard choice. Reviewers should ask whether evidence supports the inference from observed indicator to intended construct, whether the inference holds across subgroups and contexts, and whether plausible gaming or adaptation would break the link. This aligns with validity-as-argument approaches in measurement and program-evaluation practice (Messick, 1995; American Educational Research Association et al., 2014; Shadish et al., 2002).

3. Measurement Architecture and Anti-Goodhart Design

3.1 Portfolio principle

Any civilizational target becomes dangerous when a proxy becomes the target. The protocol therefore forbids single-KPI optimization. It requires portfolios, process-plus-outcome pairing, integrity indicators, uncertainty disclosure, subgroup reporting, periodic metric review, and adversarial probes. The construct remains central; indicators are imperfect instruments that support learning, not idols that replace judgment.

No portfolio architecture can make Goodhart effects impossible. Goodhart variants include regressional, extremal, causal, and adversarial failures; the protocol therefore aims to raise the cost of gaming, preserve construct visibility, detect proxy drift earlier, and require repair when indicators cease to track the underlying flourishing construct (Manheim & Garrabrant, 2018).

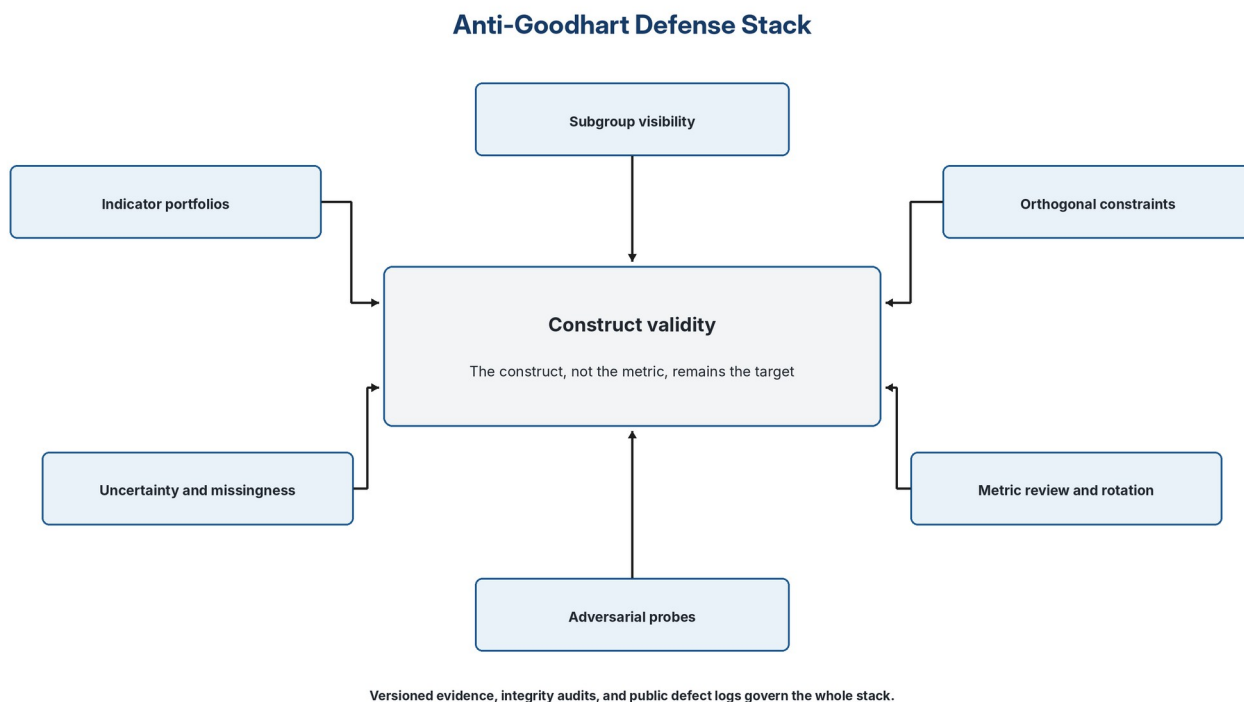


Figure 2. Anti-Goodhart defense stack. The construct remains central; indicators support it through portfolios, subgroup visibility, orthogonal constraints, integrity audits, uncertainty bands, metric review, versioned changelogs, and adversarial probes.

3.2 Three additional safeguards

- Orthogonal constraints: each optimization target must be paired with a constraint metric that detects harm in a different measurement dimension. Fraud-detection efficiency, for example, must be paired with false-positive and subgroup-harm checks.
- Decoupling reward from telemetry: metrics should guide strategy and trigger review, but should not operate as mechanical payout or punishment triggers without contextual audit.

- **Metric review and rotation:** Stable core indicators should remain available for longitudinal comparability and archival trend analysis. Rotation applies mainly to primary optimization targets, high-stakes incentive metrics, and process indicators vulnerable to institutional test-gaming. Formal depreciation equations are treated as optional research directions, not as canonical requirements in this paper.

3.3 Uncertainty and subgroup discipline

Every major indicator must report uncertainty. Objective indicators require confidence intervals or error bands where possible. Subjective or mixed indicators require evidence grades, sampling caveats, rater-bias discussion, and sensitivity checks. For rights-sensitive settings, uncertainty must affect permitted actions. Low confidence should not be used to assume compliance. When evidence is noisy and rights are at stake, the protocol requires conservative escalation, additional review, containment, or refusal under underdetermination.

All portfolios must report distributional effects. At minimum, reporting should include median, bottom quintile, worst affected protected subgroup where legally and ethically permissible, and a relevant inequality measure. If privacy rules prevent subgroup reporting, the submission must provide a privacy-preserving substitute and disclose what remains unobservable.

Data minimization is part of measurement discipline. Candidate frameworks must justify collection, retention, access control, sensitive-attribute handling, privacy-preserving subgroup analysis, and security safeguards for every indicator that relies on personal or sensitive data.

When direct subgroup reporting is not lawful or safe, the substitute method must itself be specified in the measurement card. Acceptable substitutes may include cell suppression thresholds, k-anonymity rules, differential-privacy parameters, secure multiparty analysis, synthetic-data validation, re-identification risk testing, or independent confidential review. A submission may not simply state “privacy” as a reason for hiding worst-off-group effects; it must disclose what remains unobservable and how the residual risk will be monitored.

3.4 Measurement-card requirements and bias safeguards

This handbook requires a higher scientific burden for indicators used in high-impact comparisons. Each material indicator should be accompanied by a validity argument rather than a mere definition. The card must state what inference the indicator is allowed to support, what inference it cannot support, and what would force revision, downgrade, or retirement.

Additional field	Required content
Construct-validity argument	Why the indicator supports the intended F2PHS+ construct and what evidence would falsify that inference.
Causal status	Whether the indicator is an outcome, proxy, leading indicator, process measure, integrity signal, causal lever, or mixed measure.
Measurement invariance	Whether the indicator behaves comparably across cultures, languages, institutions, protected groups, and time.
Missing-data mechanism	Whether missingness is likely random, conditionally random, or systematic, with subgroup and rights implications.
Decision relevance	Which decisions the indicator may support, and which decisions it must not support.
Burden and harm risk	Whether collecting or using the indicator creates surveillance, stigma, exclusion, administrative burden, or chilling effects.

Additional field	Required content
Retirement/ deprecation condition	When proxy drift, gaming, changed law, changed science, or empirical failure should downgrade or retire the metric.
Independent verification path	Who can audit the indicator, what they need to access, and how confidentiality or privacy limits are handled.
Decision authority boundary	What kinds of decisions this indicator may inform, what decisions it must not decide alone, and what lawful or institutional authority remains required before action.

Subjective wellbeing and meaning measures remain important because lived experience matters. They are not permission structures. They cannot override rights floors, ecological harm, coercion, or subgroup injury. Submissions using subjective measures must discuss adaptive preferences, fear-induced reporting, social desirability bias, authoritarian preference falsification, cultural response styles, and survey nonresponse.

The measurement-card requirement adapts the documentation logic of model cards and datasheets for datasets to civilizational indicators: every indicator must disclose what it measures, what it misses, how it can fail, and how it should be audited (Gebru et al., 2021; Mitchell et al., 2019).

Table 5. Measurement card template.

Field	Required content
Indicator name	Plain-language name and technical name
Construct link	Which F2PHS+ dimension and why it belongs
Definition	Exact formula, numerator, denominator, inclusion and exclusion rules
Data source	Source, owner, access, update cadence, quality issues
Uncertainty	Error bands, confidence intervals, evidence grade, rater bias, missingness
Subgroup reporting	Required disaggregation or privacy-preserving substitute
Gaming risks	How actors might manipulate the indicator
Integrity checks	Missingness flags, audit sampling, independent verification, anomaly tests
Revision history	Version, date, changes, rationale, responsible party
Privacy, security, and data minimization	Purpose limitation, retention rules, access controls, sensitive-attribute handling, privacy-preserving subgroup analysis, and security safeguards
Decision authority boundary	What kinds of decisions this indicator may inform, what decisions it must not decide alone, and what lawful or institutional authority remains required before action.

For cross-country, cross-cultural, or multilingual comparison, measurement cards should also state whether measurement invariance has been evaluated. Where suitable, reviewers may request differential item functioning checks, multi-group factor analysis, alignment methods, translation/back-translation review, or explicit non-comparability warnings. Early-warning indicators should be treated as risk signals rather than causal proof unless a causal identification strategy, triangulated mechanism, or strong domain evidence is declared. Planetary-boundary and resilience indicators also carry control-variable uncertainty, threshold uncertainty, and interaction uncertainty; these uncertainties should be disclosed rather than hidden behind a single ecological score.

3.5 Non-documentary evidence measurement card

For oral, observed, customary, community, translated, or otherwise non-documentary evidence, the measurement card additionally records source authority, consent basis, purpose limit, custody, disclosure restrictions, translation or interpretation, verification, triangulation, correction, challenge route, compensation, withdrawal, retention and deletion, and data sovereignty. Reviewers assess function and challengeability rather than document volume, visual polish, legal drafting, or computational form.

4. F2PHS+ reference architecture

Table 6. F2PHS+ reference architecture. The long-form layout makes the expanded structure beneath the communication layer readable without implying that every property can be reduced to one indicator.

Property	Role	Reference content
Free	Floor	Bodily integrity, liberty, due process, privacy, and non-coercion.
Free	Field	Agency, voice, participation, and informational autonomy.
Free	Horizon	Freedom preserved during emergencies, automation, surveillance, and institutional drift.
Free	Integrity guardrail	Notice, appeal access, surveillance audit, and complaint-channel integrity.
Free	Indicator families	Civil liberties, rights-in-practice gaps, appeal timeliness, privacy incidents.
Free	Prohibited misuse	Calling a society free when fear, automation, or opacity prevents meaningful challenge.
Fair	Floor	Non-discrimination, equal protection, fair procedure, and anti-domination.
Fair	Field	Equitable opportunity, administrative fairness, anti-corruption, and burden distribution.
Fair	Horizon	Reduced structural exclusion and long-term institutional legitimacy.

Property	Role	Reference content
Fair	Integrity guardrail	Subgroup audit, procurement integrity, and selective-enforcement checks.
Fair	Indicator families	Disparity ratios, court access, corruption controls, discrimination complaints.
Fair	Prohibited misuse	Averaging away harms to protected or worst-off groups.
Prosperous	Floor	Basic needs, subsistence, housing security, and health access where rights-relevant.
Prosperous	Field	Material security, health, education, work, social protection, and resilience.
Prosperous	Horizon	Shock resilience, poverty persistence, and capability expansion over time.
Prosperous	Integrity guardrail	Benefit-denial audit, false-positive review, and survey integrity.
Prosperous	Indicator families	Poverty gap, MPI, food insecurity, avoidable mortality, learning outcomes.
Prosperous	Prohibited misuse	Treating GDP, efficiency, or aggregate output as sufficient evidence of prosperity.
Harmonious	Floor	Protection from violence, repression, intimidation, and forced conformity.
Harmonious	Field	Social trust, relationships, constructive pluralism, conflict-resolution capacity, and meaning.
Harmonious	Horizon	Polarization, organized violence, institutional breakdown, and intergroup dehumanization.
Harmonious	Integrity guardrail	Dissent protection, media integrity, bot/manipulation signals, and grievance-channel review.
Harmonious	Indicator families	Homicide, UCDP conflict, displacement, trust, mediation access, affective polarization.

Property	Role	Reference content
Harmonious	Prohibited misuse	Equating harmony with obedience, silence, censorship, or visible order.
Sustainable	Floor	Ecological integrity where life, health, subsistence, or community survival is directly implicated.
Sustainable	Field	Environmental quality, pollution reduction, and local ecosystem health.
Sustainable	Horizon	Planetary boundaries, climate stability, biodiversity, and intergenerational resilience.
Sustainable	Integrity guardrail	Greenwashing audit, offset integrity, and boundary-data verification.
Sustainable	Indicator families	Emissions, PM2.5, freshwater stress, land use, biodiversity proxies, adaptation capacity.
Sustainable	Prohibited misuse	Using offsets or aggregate environmental scores to hide local or irreversible harm.

This dictionary is provisional for challenge rounds. It is canonical only for comparability under this protocol family, not as a final theory of human or civilizational value. Candidate frameworks may rename dimensions or use richer ontologies, but they must map their constructs to the protected-floor, field, horizon, integrity, and indicator roles above or justify a principled alternative. No top-level property may be represented by a single KPI.

5. Worked measurement-card examples

Worked measurement-card examples. These are illustrative minimums for a public challenge packet; final cards must be adapted to jurisdiction, domain, authority, and data availability.

Table 7. Worked measurement-card examples in long form.

Measurement card	Required element	Illustrative content
Appeal access and timeliness	Construct link	Free/Fair floor and process indicator.
Appeal access and timeliness	Definition	Share of affected persons receiving notice, explanation, accessible appeal, and decision within a declared timeframe.
Appeal access and timeliness	Uncertainty and subgroups	Report median, bottom quintile, protected groups where lawful, disability/language access, missingness, and confidence limits.

Measurement card	Required element	Illustrative content
Appeal access and timeliness	Gaming and integrity checks	Audit random cases; compare appeal success rates, abandoned appeals, inaccessible notices, and complaint suppression.
Appeal access and timeliness	Revision trigger	Delay, access gap, or appeal-success reversal exceeds a predeclared threshold.
Missing-document denial risk	Construct link	Fair/Prosperous rights-sensitive integrity indicator.
Missing-document denial risk	Definition	Rate at which missing documentation leads to denial, delay, or manual review, separated from confirmed ineligibility.
Missing-document denial risk	Uncertainty and subgroups	Disaggregate by protected or worst-off groups, or use confidential/privacy-preserving review.
Missing-document denial risk	Gaming and integrity checks	Detect documentation-burden laundering, subgroup missingness, and fraud-flag overreach.
Missing-document denial risk	Revision trigger	Missingness predicts adverse action more strongly than eligibility evidence or diverges materially by subgroup.
Eviction-after-contact outcome	Construct link	Prosperous field and floor-risk outcome.
Eviction-after-contact outcome	Definition	Share of applicants with housing-risk contact who experience eviction, homelessness, or unresolved arrears within a defined follow-up window.
Eviction-after-contact outcome	Uncertainty and subgroups	Use survival or time-to-event logic where appropriate; report uncertainty and data-linkage gaps.
Eviction-after-contact outcome	Gaming and integrity checks	Test whether apparent fraud efficiency increases downstream housing harm.
Eviction-after-contact outcome	Revision trigger	Emergency-stabilization failure or subgroup divergence exceeds the declared threshold.
Protected dissent and grievance-channel integrity	Construct link	Harmonious integrity and anti-conformity safeguard.

Measurement card	Required element	Illustrative content
Protected dissent and grievance-channel integrity	Definition	Presence, accessibility, safety, and responsiveness of complaint, protest, media, and deliberation channels.
Protected dissent and grievance-channel integrity	Uncertainty and subgroups	Report under-sampling, fear-response risk, retaliation reports, and journalist/opposition detention proxies where relevant.
Protected dissent and grievance-channel integrity	Gaming and integrity checks	Compare trust scores against repression, censorship, complaint suppression, and visible-conflict measures.
Protected dissent and grievance-channel integrity	Revision trigger	Trust rises while dissent channels narrow or retaliation indicators increase.
Boundary-aligned ecological stress	Construct link	Sustainable horizon and local floor where life support is implicated.
Boundary-aligned ecological stress	Definition	Local or sectoral indicator linked to a planetary-boundary or safe-operating-space family, such as freshwater stress, PM2.5, emissions trajectory, biodiversity proxy, or land conversion.
Boundary-aligned ecological stress	Uncertainty and subgroups	Report model uncertainty, local hotspots, affected communities, lag effects, and boundary mismatch.
Boundary-aligned ecological stress	Gaming and integrity checks	Audit greenwashing, offsets, scope shifting, and aggregate masking of local collapse.
Boundary-aligned ecological stress	Revision trigger	Boundary trajectory worsens, offset integrity fails, or local life-support risk becomes rights-relevant.

6. Enhanced measurement-card fields

The following fields extend the measurement-card template for high-impact uses. They are recommended for Core-tier challenges and required for Full-tier validation unless a public justification explains why a field is inapplicable.

Table 8. Enhanced measurement-card fields expose validity, burden, verification, and authority risks.

Field	Reviewer question	Failure signal
Construct-validity argument	Why should this indicator support the intended construct?	Indicator is convenient or prestigious but not tied to the claimed construct.

Field	Reviewer question	Failure signal
Causal status	Is the indicator an outcome, proxy, leading indicator, process measure, integrity signal, or causal lever?	Indicator is used to justify action beyond its inferential role.
Measurement invariance	Does the indicator behave comparably across cultures, languages, institutions, groups, and time?	Observed differences may reflect measurement artifact rather than real differences.
Missing-data mechanism	Is missingness random, conditionally random, or systematic?	Missingness tracks vulnerability, exclusion, fear, instability, or administrative burden.
Decision relevance	What decisions may this indicator support or not support?	Indicator is used for denial, ranking, or coercion beyond its design scope.
Burden and harm risk	Can collecting or using the indicator create surveillance, stigma, exclusion, or chilling effects?	Measurement itself harms the group being measured.
Retirement/deprecation condition	What would make this indicator stale, gamed, unlawful, invalid, or misleading?	Metric persists after proxy drift or observed failure.
Independent verification path	Who can audit it, under what access rules, and with what privacy safeguards?	No independent route exists to inspect the data, assumptions, or transformation.
Decision authority boundary	What decision authority is being claimed?	Indicator is used to deny, rank, coerce, authorize, or certify beyond its permitted evidentiary role.

7. Evidence maturity and confidence

Every rating or indicator claim reports evidence maturity, evidence mode, and reviewer confidence separately. `evidence_maturity_level` uses L0-L6. `evidence_mode` records how evidence was produced, such as conceptual analysis, synthetic replay, historical backtest, independent replay, shadow mode, pilot, live observation, or experiment. The record also states `maturity_subject`, `maturity_rationale`, and `evidence_pointer`. Reviewer confidence is high, moderate, low, or unresolved. None of these fields is added arithmetically to an ordinal category.

7.0 Common Stage B anchor ladder

All twelve dimensions instantiate the same pattern: 0 absent/defeating; 1 nominal/weak; 2 operative/inspectable; 3 independently challenged; 4 demonstrated/corrected. Dimension-specific Appendix F anchors control. A shared ladder does not establish one latent scale.

7.1 Ordinal category interpretation and calibration

The core paper owns the twelve Stage B dimensions and their ordered 0-4 descriptors. This handbook governs evidence interpretation and calibration. Reviewers must not sum, average, subtract, or assign decimal materiality thresholds to the category labels. Category evidence is reported alongside evidence maturity level, evidence mode, maturity subject and rationale, evidence pointers, reviewer confidence, and material disagreement.

Calibration cases should test boundary conditions between adjacent categories, not merely obvious 0-versus-4 examples. Each case should identify the decisive evidence, plausible alternative interpretation, affected subgroup or rights issue, and what additional evidence would change the category. Low agreement at one boundary is evidence that the anchor, evidence requirement, or reviewer training may be defective.

Agreement methods must fit the scale. With expected concentration in middle ordinal categories, Gwet AC2 is a defensible primary coefficient and ordinal Krippendorff alpha a useful cross-check; weighted kappa is a secondary comparator whose weighting and prevalence sensitivity must be disclosed. ICC is reserved for justified interval or continuous indicators. The protocol does not prescribe a universal numerical reliability threshold. Targets and consequences must be preregistered for the specific pilot (Gwet, 2014; Krippendorff, 2018; Liddell & Kruschke, 2018).

Reliability must be reported separately for each C0-C5 criterion, each Stage B dimension, the final Stage A outcome, and the final Stage B outcome. A mean that mixes criterion categories with categorical final decisions is invalid. Phase 0 estimates are exploratory; a later confirmatory design uses simulation-based precision or decision-error planning informed by observed prevalence and dependence. Criterion order is randomized or balanced, and global-impression probes are recorded before and after rating so halo and order effects can be estimated rather than mistaken for shared criterion interpretation. Ratings materially suggested or generated by a common AI system are analyzed in a separate condition and are not treated as independent human ratings.

7.2 Not applicable, unknown, and zero

A category of 0 means the dimension is applicable and the required function is absent, contradictory, or materially failing. It must not encode **unknown**, **not reviewed**, or **not applicable**. Machine records and public reports must preserve these states separately with a rationale. A profile with a material unknown cannot support dominance if the missing dimension could reverse the relation. Core Table 8M controls the resulting public outcome: a valid but evidentially unresolved comparison is **non_decisive**; invalid comparison prerequisites require **refusal_to_compare**; adequately evidenced crossing profiles are **incomparable**.

For dimensions fully linked to a Stage A floor, category 0 is presumptively incompatible with continuing Stage B eligibility. Reviewers return the candidate to Stage A when later evidence establishes a material C0–C5 defect. Category 1 on a floor-linked dimension is reportable only with a written explanation of why the weakness did not reach Stage A materiality. Partly linked dimensions require an explicit **floor_linkage** record and the same analysis for the linked portion.

Reviewer confidence on a **not_applicable** record attaches to the applicability rationale, not to a nonexistent category. An applicability mismatch or category disagreement is reversal-capable when substituting an evidence-supported alternative changes the partial-order relation; the comparison record must preserve and rerun each defended alternative.

7.3 Assisted and differential evidence pathways

Measurement integrity does not require every architecture to produce the same artifact form. A measurement card may bind evidence from structured data, reasons records, observation, testimony, oral or translated deliberation, customary practice, community records, protected administrative data, or computational logs when the host can reconstruct provenance, scope, construct link, uncertainty, distribution, and challengeability.

Assistance changes the form of evidence, not the evidentiary standard. The card records the original form, assistance provider, translation or transformation, candidate approval, information loss, confidentiality constraints, and whether the assisted representation could alter interpretation. Reviewers assess polished presentation separately from construct support. No candidate gains a higher category merely because it can purchase more elaborate documentation.

Pilots compare missingness, reviewer time, confidence, disagreement, and category outcomes across evidence pathways and resource groups. Systematic disadvantage for community, customary, low-resource, disabled, or non-dominant-language entrants is evidence of C1-D05 and may require redesign or refusal.

7.4 Calibration sample planning, differential performance, and burden

A measurement calibration plan must identify the subject of each maturity claim, the evidence mode, the architecture families represented, reviewer structure, affected-party participation, and the precision required for the intended inference. It reports criterion- and dimension-level agreement, differential missingness and burden by architecture type, the effect of assisted-dossier support, and whether particular evidence forms systematically generate lower confidence or Indeterminate outcomes.

Formal power calculations are used only when the estimand, sampling model, effect size, and dependence structure are defensible. Otherwise, the study uses precision-based planning and declares target uncertainty intervals, minimum informative case diversity, reviewer count, and stopping rules. Exploratory factor or redundancy analysis of AF-SB12 remains hypothesis-generating until sample size and ordinal-model assumptions are justified.

Burden is a measurement-integrity variable rather than an administrative afterthought. The calibration record includes preparation time, review and adjudication time, translation and accessibility support, assistance hours and cost, withdrawals, missing evidence caused by resource limits, and whether assistance materially changed a finding. Persistent resource-linked missingness, withdrawal, or Indeterminate outcomes trigger redesign of the evidence pathway rather than automatic attribution to candidate opacity.

8. Indicator retirement and construct revision

Indicators should be retired or redesigned when they lose construct validity, create unacceptable burden, become gameable, conceal subgroup harm, depend on unstable access, violate privacy, or cease to inform action. A construct vocabulary should be revised when affected parties or diverse candidates demonstrate systematic omission or architecture bias. Revision must preserve a public changelog and explain whether historical comparisons remain valid.

AF-SB12-v5.6 dimension-set and AF-SB-RELATION-v6.0 governance

The twelve Stage B dimensions remain the provisional component AF-SB12-v5.6 because v6.2 does not add, merge, split, retire, or reorder them. They cover protected floors and power; evidence and inspectability; learning and anti-gaming; and horizon, feasibility, and institutional fit. Calibration tests overlap, omissions, burden, architecture bias, and dimension-level reliability. A change to the dimension set requires a new component version, construct map, backward-compatibility note, benchmark replay, and public evidence under C3-D07.

Version 6.2 preserves the separately versioned relation component, AF-SB-RELATION-v6.0. Reviewers compute the base partial-order relation and replay every evidence-supported alternative category. If all relations remain one-direction dominance, dominance is invariant. If all remain crossing, incomparability is invariant. If relations change, include a tie, or remain unresolved under valid prerequisites, the outcome is non-decisive. Invalid prerequisites produce refusal to compare. This symmetric rule prevents peripheral disagreement from erasing a stable crossing while preventing fragile dominance from being overstated.

A category is floor-linked unreachable when its descriptor necessarily entails a material Stage A defect under the same scope. Rights robustness category 0 and Refusal quality category 0 are not stable Stage B values; they route to Stage A. Category 1 on either requires a written materiality rationale. Other dimensions route back when the evidence establishes a material C0-C5 defect.

Calibration reports Stage A outcomes, eligible pairs, base dominance, invariant dominance, invariant incomparability, non-decisiveness, refusal, ties, relation changes, and reason-coded empty sets. The v6.2 supplement provides five-category, range-restricted, mixed, and correlated structural sensitivities. These are model diagnostics, not forecasts.

Stage A can restrict the observed category range, especially on dimensions close to C0-C5. This can raise base-profile dominance while simultaneously making dominance fragile when it rests on one adjacent category. The first external studies preregister competing hypotheses and report the actual relation set. They do not assume in advance that dominance, incomparability, or non-decisiveness must prevail.

The first empirical calibration also tests whether dimensions load onto the four functional groups. Factor or latent-structure analysis is exploratory until ordinal-model assumptions and sample size are justified. Group summaries remain explanatory and cannot replace the dimension-level partial order without separate normative justification.

A not_applicable judgment requires a rationale, scope binding, and reviewer confirmation. A potentially resolvable mismatch produces non-decisiveness; inability to establish a valid common scope produces refusal to compare.

Burden as a measurement and validity construct

Burden is multidimensional and distributional. The minimum record separates candidate preparation, reviewer assessment, adjudication, affected-party participation, access and translation, evidence security, appeal, direct financial cost, uncompensated time, delay, abandonment, and subjective burden. Means cannot hide role-specific or subgroup concentration. A study must report medians, ranges, tails, missingness, and architecture-, language-, disability-, resource-, and documentation-intensity differences where feasible.

Feasibility is conjunctive: all required capacity and access conditions must be met. The v6.2 aggregation workbook and reference engine deliberately include a false-green case in which per-case hours are acceptable while reviewer count, calendar, and interpretation fail. The correct result is not feasible.

External construct and architecture-fairness validation

Validation proceeds through content review, response-process evidence, inter-reviewer reliability, internal-structure analysis where appropriate, relations to external variables, decision utility, consequences, and measurement invariance. Because AF contains ordinal and normative constructs, no single coefficient establishes validity. The external study must include functionally equivalent architectures with different artifact forms and must estimate differential false clearance, false rejection, indeterminacy, burden, and appeal reversal.

The audit-theatre arm crosses document polish with substantive safeguard quality. A valid instrument should reject or question polished but deficient records and should not penalize substantively adequate low-polish records after appropriate support. Ablation tests whether each dimension earns its burden; decision-value analysis asks whether the added information improves decisions or merely produces more documentation.

AF v6.2 synchronization note

This artifact implements the v6.2 three-axis record (decision_status, evidence_state, review_state), the non-erasure invariant, perturbation-replay anti-inflation and anti-suppression controls, and the separation of raw agreement, reliability, adjudication, and external validity. These controls clarify the existing protocol and do not confer legal, moral, procurement, or deployment authority.

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